

✓ PYTHON FOR DATA ANALYSIS WITH AI ASSISTANCE

SKILLVERSE – 2026

Skill Development Cell

Duration: 15 Sessions × 2 Hours = 30 Hours

Number of Students: 40

Primary Platform: Google Colab

Alternative Notebook Environment: Jupyter Notebook

1. Course Overview

Python for Data Analysis with AI Assistance is a practical skill-development course designed to introduce students to the process of analysing real-world data using Python.

The course focuses on **Data Analysis with Python rather than teaching Python programming from the beginning**. Only the Python concepts required to understand and perform data analysis will be introduced when needed.

The course is designed for students from different academic backgrounds, including:

- Computer Science
- Data Science
- Commerce
- Biology
- Other disciplines interested in working with data

The course follows a **hands-on, dataset-oriented approach**, where students learn concepts through practical analysis of real datasets.

3. Why Data Analysis?

Data is generated in almost every field such as:

- Business
- Banking and Finance

- Healthcare
- Education
- E-commerce
- Transportation
- Biology and Life Sciences
- Marketing
- Social Media

Raw data by itself may not provide useful information.

Data Analysis helps us understand data and extract meaningful information from it.

Basic Data Analysis Workflow

Collect → Understand → Clean → Analyse → Visualize → Interpret → Communicate

Through this process, meaningful patterns, trends and insights can be obtained from data.

4. Why Python for Data Analysis?

Python is widely used for Data Analysis because it provides powerful libraries and tools for handling numerical, tabular and visual data.

In this course, we will mainly work with the following libraries:

NumPy

Used for numerical and array-based operations.

Pandas

Used for handling, cleaning, transforming and analysing tabular data.

Matplotlib

Used for creating data visualizations.

Seaborn

Used for statistical and attractive data visualizations.

Python also provides an extensive ecosystem for further Data Science and Machine Learning applications.

In this course, our focus is not on learning Python programming separately. Our focus is on using Python to perform Data Analysis.

5. Python Prerequisites for Data Analysis

This course does **not** aim to teach Python programming from the beginning.

However, students need to understand a few basic Python concepts required for Data Analysis.

These concepts will be introduced briefly wherever required:

- Variables
- Numbers and strings
- Lists
- Dictionaries
- Basic operators
- Functions
- Conditions
- Basic loops
- Importing libraries
- Calling functions

Important

The emphasis will remain on **using Python to analyse data**, not on learning general-purpose Python programming.

6. Learning Environment

Google Colab

Students will primarily use **Google Colab** because it provides:

- Browser-based Python environment
- No local installation required
- Easy execution of Python code
- Notebook-based learning
- Easy sharing of notebooks
- Integration with Google Drive

- Suitable environment for practical Data Analysis

Jupyter Notebook

Students will also be introduced to **Jupyter Notebook** so that they understand the commonly used notebook-based Data Analysis environment.

Course Preference

Google Colab will be the primary platform for all student coding activities.

7. Datasets Used

Dataset 1 — Sample Superstore

The **Sample Superstore dataset** will be the primary teaching dataset.

It will be used to demonstrate:

- Data inspection
- Data selection
- Data filtering
- Data cleaning
- Numerical analysis
- GroupBy
- Exploratory Data Analysis
- Visualization
- Business-oriented questions
- Interpretation of results

Students will repeatedly work with this dataset while learning new techniques.

<https://www.kaggle.com/datasets/himanshuuike/superstore-sales-dataset>

Dataset 2 — Airline Flights

The **Airline Flights dataset** will mainly be used for independent practice and project work.

Students will apply the techniques learned from the Superstore dataset to a different real-world context.

Possible areas include:

- Airline-wise analysis
- Flight delays
- Time-based analysis
- Airport analysis
- Numerical analysis
- Categorical analysis
- Visualization
- Exploratory Data Analysis
- Business/operational insights

<https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data>

8. Role of AI Assistance

AI will be integrated throughout the course as a **learning and productivity assistant**.

Students will learn to use AI for:

- Understanding Data Analysis concepts
- Explaining Python, Pandas and NumPy functions
- Generating code
- Modifying code
- Debugging errors
- Suggesting analytical questions
- Selecting appropriate visualizations
- Interpreting results
- Improving code
- Generating ideas for projects

Important Principle

AI-generated code must be understood, executed and verified by the student.

Students should not simply copy AI-generated answers.

The purpose of AI assistance is to help students **learn, analyse, question and improve**, not to replace their own understanding.

9. Overall Data Analysis Workflow

Throughout the course, students will gradually learn the following workflow:

Raw Dataset

Load Data

Understand Data

Inspect Data

Clean Data

Select / Filter Data

Transform Data

Perform Numerical Analysis

Group and Aggregate

Exploratory Data Analysis

Visualize Data

Interpret Results

Generate Insights

Communicate Findings

This workflow will be applied first to the **Sample Superstore dataset** and later to the **Airline Flights dataset**.

10. Course Learning Outcomes

By the end of the course, students should be able to:

1. Explain the basic Data Analysis workflow.
2. Work with Google Colab and Jupyter Notebook.
3. Load and inspect datasets using Pandas.
4. Perform numerical operations using NumPy.
5. Select and filter data using Pandas.
6. Clean basic real-world datasets.
7. Perform grouping and aggregation.
8. Conduct basic Exploratory Data Analysis.
9. Create appropriate visualizations using Matplotlib and Seaborn.

- 10. Interpret analytical results and communicate insights.
- 11. Use AI tools responsibly for Data Analysis tasks.
- 12. Apply the complete Data Analysis workflow to a new dataset.
- 13. Develop and present a small Data Analysis project.

11. Assessment Structure

Assessment	Purpose
Pre-Assessment	Identify students' existing knowledge before the course
Daily Exercises	Continuous practical learning
Day-7 Assessment	Evaluate learning from the first half of the course
Final Project	Apply the complete Data Analysis workflow
Post-Assessment	Measure learning after completing the course
Student Feedback	Collect feedback about the course

12. 15-Day Course Workplan

Total Duration: 15 Sessions × 2 Hours = 30 Hours

Day	Date	Venue	Session Type	Main Topic	
1	Aug 12	Day Lab 2	Lab	Introduction to Data Analysis & Google Colab	Data Analysis w
2	Aug 13	B1 Garden Block	Class	NumPy & Pandas GroupBy	NumPy arrays, i
3	Aug 17	Day Lab 2	Lab	Pandas Data Selection & Filtering	Column selectio
4	Aug 18	B1 Garden Block	Class	Data Cleaning & Preparation	Missing values,
5	Aug 19	Day Lab 2	Lab	Data Cleaning Hands-on	Missing-value h
6	Aug 20	B1 Garden Block	Class	Exploratory Data Analysis	Univariate anal
7	Aug 24	Day Lab 2	Assessment	7-Day Assessment + EDA Practice	NumPy, Pandas
8	Aug 25	B1 Garden Block	Class	Data Visualization with Matplotlib	Importance of v
9	Aug 28	Day Lab 2	Lab	Seaborn & Visualization Practice	countplot), (t
10	Aug 29	B1 Garden Block	Class	Complete EDA Workflow	Data questions,
11	Aug 31	Day Lab 2	Lab	Airline Flights Dataset	Dataset unders
12	Sep 1	B1 Garden Block	Class	Airline EDA & Visualization	Airline-wise an
13	Sep 2	Day Lab 2	Lab	AI-Assisted Data Analysis	AI for explanati
14	Sep 3	B1 Garden Block	Class	Final Project Development	Problem definit
15	Sep 7	Day Lab 2	Final Evaluation	Final Project + Post-Assessment	Project present

> DAY 1 — Introduction to Data Analysis with Python

Date: August 12, 2026

Venue: Day Lab 2

Duration: 2 Hours

Session Type: Practical / Lab

Today's Focus

- Understanding Data Analysis
- Why Python is used for Data Analysis
- Basic Python concepts required for Data Analysis
- Google Colab
- Jupyter Notebook
- Notebook structure
- Markdown and Code cells
- Introduction to NumPy and Pandas
- Loading the Sample Superstore dataset
- First Data Inspection

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> DAY 2 — NumPy and Pandas GroupBy for Data Analysis

Course: Python for Data Analysis with AI Assistance **Duration:** 2 Hours **Session:** Day 2
Dataset: Sample Superstore **Practice Dataset:** Airline Flights **Platform:** Google Colab

Today's Topics

1. Introduction to NumPy for Data Analysis
2. NumPy arrays
3. Numerical operations
4. NumPy statistical functions
5. Pandas aggregation
6. `sum()`, `mean()`, `min()`, `max()`, `std()`
7. Introduction to `groupby()`
8. Group-wise analysis
9. Multiple aggregations using `agg()`
10. Sorting grouped results

11. Business questions using GroupBy
12. AI-assisted Data Analysis
13. Home Task — Airline Flights Dataset

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